A Real Time Subband Implementation of an Active Noise Control System for Snoring Reduction

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***Abstract*—In this paper, a real time implementation of an active noise control (ANC) system for snoring cancellation is presented. This method is based on an online secondary path estimation procedure and includes a subband adaptive filtering (SAF) structure for the primary path estimation. The introduction of the SAF approach allows to improve the performances on the estimation of both primary and secondary path in terms of convergence rate. The algorithm has been compared with the state of the art through several experiments that have proved the effectiveness of the proposed system in real applications.**

***Index Terms*—active noise control, subband adaptive filtering, filters bank, snoring reduction**

I. Introduction

The increasing interest in active noise control (ANC) systems has opened the way to several applications. In the recent years, research has focused on the employment of ANC techniques for snoring reduction. This particular noise can have a sound pressure of more than 90 dB and it can impact to bed partner life in relation to work productivity or hearing loss [1] [2]. Recent researches have observed that the snoring signal has common characteristics to the vocal signal. Both types of signal show a fundamental frequency and some high order harmonics [3]. The whole snoring period is composed by two intervals: the inspiration and the expiration. Most of the signal power is located in the low frequencies range during these two phases. Therefore, an efficient noise reduction system capable to work at low frequencies is required, because the energy of the signal to be removed is located around 200 Hz. Regarding the literature, the problem of snoring reduction is solved in different ways. Referring to passive techniques, earplugs or particular pillow can be employed [4]. However, sleeping with earplugs or other devices might be uncomfortable for the user. In addition, passive systems are usually ineffective at low frequencies. For this reasons, active solutions are more suitable for this task, thanks to their capacity to reduce the low frequency noise. ANC techniques require filters capable of tracking the modifications of the acoustic path. For the filters design, several methods can be found in the literature. Among them, least mean square (LMS) algorithm is the most implemented. The ANC systems consider a secondary source that generates an artificial anti-noise signal for the attenuation of the primary noise. In this way, the secondary path between the source and the error microphone must be determined. Taking into consideration the secondary path, the use of filtered-x least mean square (FxLMS) have been proposed in the literature [5] [6] to solve the ANC problem. Considering a standard feedforward structure using the FxLMS algorithm, the secondary path impulse response is known and the filter is adapted with the residual noise, recorded by the error microphone, using the LMS algorithm. However, the secondary path can vary significantly in time due to environmental conditions or loudspeakers and microphones damage leading to an instability of the algorithm. To avoid this problem, the FxLMS structure can be modified with the implementation of an adaptive filter for secondary path. Different approaches have been developed for online secondary path modeling. The first approach proposed by [7] is based on injection of additional random noise, uncorrelated with the primary noise, into the ANC system to model the secondary path, while another method that avoid the injection of additional noise is proposed by [8]. These two approaches have been compared in [9], by concluding that the approach proposed by Eriksson (i.e., [7]) presents an improvement with relation to the algorithm proposed by [8] in particular considering the performance of primary path estimation.

In this context, two different approaches have been introduced by Bao et al. [10] and Kuo et al. [11] that improve the performance of the algorithm presented by Eriksson mitigating the perturbation introduces by the injection of additional noise. These methods add an additional adaptation filter to avoid the perturbation in the primary path estimation due to the injected noise. In particular, in [10] the convergence rate is improved but the additional noise affects the convergence rate of the added filter while in [11], an adaptive prediction error

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filter is used to attenuate the interference of the additional noise. However, for both algorithms the effect of the additional noise on the adaptive filter of primary path are not evaluated. Zhang et al. go into this aspect and in [12] introduce a new structure where three different adaptive filters have been added to eliminate the perturbation of the additional noise.

In this paper, a real time implementation of the algorithm proposed in [13] is presented. In particular, the implemented structure is capable of improving the primary and secondary path estimation in terms of convergence rate and error. The algorithm proposed in [13] uses the delayless subband algorithm proposed by [14] to increase performance of primary path estimation and convergence rate while for the secondary path estimation the approach proposed by Zhang et al. are considered. The efficient algorithm of the primary path adaptation improves the performance of the whole structure since the filters are dependent on each adaptive filters. In the presented paper, the performance of the algorithm has been evaluated with several computer simulations and real applications introducing a real time implementation of the proposed system.

The paper is organized as follows. Section II describes the proposed algorithm of [13]. Section III describes the real time implementation, the setup and the validation test of the algorithm. Section IV-B explains the real time experiments and Section V gives some concluding remarks.

II. Algorithm Description

The approach of [13] is shown in Figure 1. The secondary path estimation is based on the algorithm proposed in [12] with the introduction of delayless subband algorithm presented on [14], that aims to enhance the efficiency of the whole ANC structure.

A. Secondary path estimation

The structure of the algorithm is based on feedforward FxNLMS to eliminate the perturbation of secondary path by filtering the reference signal $x(n)$ with an online estimation of secondary path $\hat{s}(n)$. The approach of [12] has been developed to remove the perturbation due to additional noise injected to estimate the secondary path. The uncorrelated noise used to adapt the secondary path has been added to the output of the controller and this signal constitutes a disturbance for the primary path estimation. To resolve the problem, the scheme of [12] computes the signal error for primary path adaptation as follows:

$$e'(n) = e(n) - \hat{s}(n) * v(n), \tag{1}$$

where $\hat{s}(n)$ is the measured secondary path, $v(n)$ is the additional noise used for the secondary path adaptation and the error $e(n)$ is determined as:

$$e(n) = d(n) - s(n) * y(n) + s(n) * v(n), \tag{2}$$

where $y(n)$ is the anti-noise from the secondary source, $d(n)$ is the noise to cancel and $s(n)$ represents the physical path from the secondary source to the error microphone. This algorithm allows to achieve the perfect cancellation $e'(n) \approx 0$ when the measured secondary path is $\hat{s}(n) \approx s(n)$ and the perturbation of additional noise is eliminated.

The adaptive algorithm of primary path employed as error the signal $e'(n)$ and the adaptive algorithm of $h(n)$ used this signal as input as shown in the following equation:

$$w(n+1) = w(n) + \mu_w x'(n) e'(n), \tag{3}$$

$$h(n+1) = h(n) + \mu_h x(n)[e'(n) - z(n)], \tag{4}$$

where $z(n)$ is the convolution between the filter $h(n)$ and the reference signal $x(n)$, $x'(n)$ is the reference signal filtered by $\hat{s}(n)$ and μ_w and μ_h are the step size of the filters $w(n)$ and $h(n)$, respectively. The signals $v(n)$ and $e_s(n)$ are used as input signal and error signal for the adaptation of the secondary path, as follows:

$$\hat{s}(n+1) = \hat{s}(n) + \mu_s v(n) e_s(n), \tag{5}$$

where μ_s is the step size of the filter $\hat{s}(n)$ and $e_s(n)$ is computed as follows:

$$e_s(n) = g(n) - \hat{u}(n), \tag{6}$$

where $g(n) = e(n) - z(n)$ and $\hat{u}(n)$ is injected noise filtered by $\hat{s}(n)$.

B. Primary path estimation

The system proposed by [12] has been modified introducing a delayless subband adaptive filtering (SAF) algorithm for the estimation of primary path. The application of SAF theory for the primary path estimation improves the convergence rate for primary path estimation and as a results of the dependence of online secondary path estimation with primary path estimation this algorithm increases the convergence rate of the entire system. In fact, the algorithm of [12] is called cross-update due to the dependence of the three adaptive filter. In Figure 1 is shown the delayless SAF structure derived from [14]. The signal $e'(n)$, calculated using the Equation (1) and the reference signal $x(n)$ filtered by the estimation of secondary path are filtered by the analysis filter-bank $\mathbf{H}(z)$, as follows:

$$\mathbf{H}(z) = [H_0(z), H_1(z), ..., H_{M-1}(z)]^T, \tag{7}$$

where $H_k(z)$ represents the $k-th$ bandpass filter with $k = 0, ..., M-1$ and M the number of subbands.

The adaptation algorithm for the $k-th$ subband weights of the primary path is described by the following equation:

$$\mathbf{w}_k^{SAF}(n+1) = \mathbf{w}_k^{SAF}(n) + \mu_w \frac{\mathbf{x}_k'^*(n) e_k'(n)}{\alpha + ||\mathbf{x}_k'(n)||^2}, \tag{8}$$

where $x_k'(n)$ and $e_k'(n)$ are the subband signals respectively of reference signal and error signal, $*$ indicates the complex conjugation, μ_w is the step size and α is a small number used to prevent a zero in the denominator.

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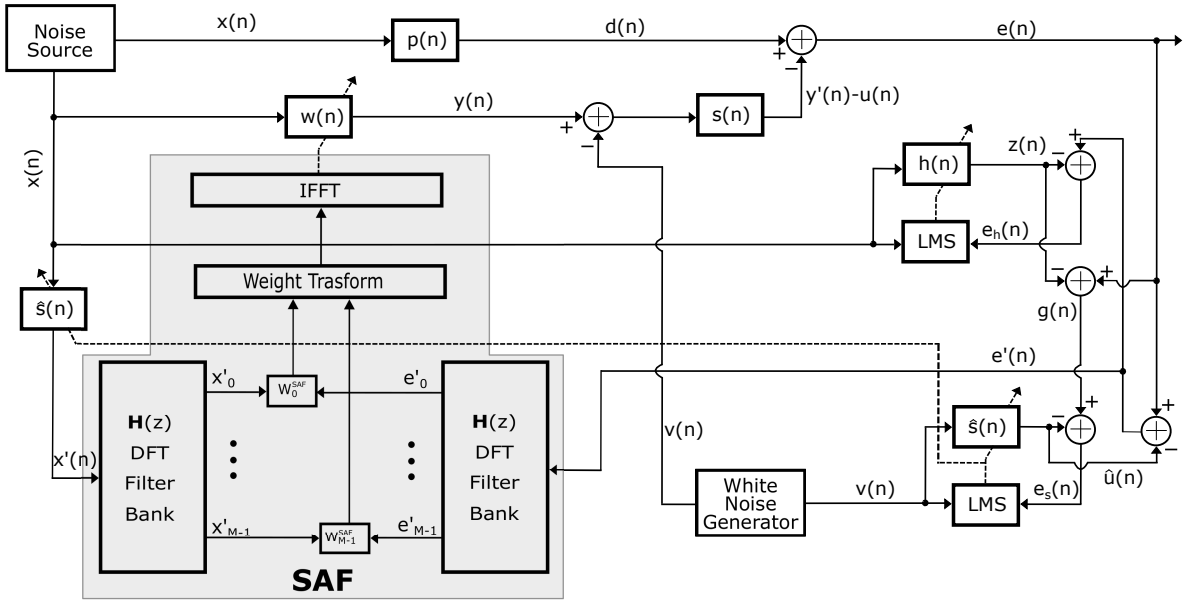


Fig. 1. Structure of the proposed ANC approach with delayless subband algorithm and secondary path estimation with injected noise.

The weights of the subband adaptive filters are stacked to compute the fullband adaptive filter following these steps:

- the first step is to compute the FFT of (N/D)-point for each subband weights, where N represents the length of the fullband filter and $D = M/2$ the decimation factor;
- the complex bin have to be stacked to calculate the first half of the vector of the fullband filter;
- the second half of the filter is filled with the complex conjugate of the first half in reverse order;
- to obtain the fullband filter a N-point IFFT is applied to the vector.

III. Real-Time Implementation

The real-time implementation has been developed using the software NU-Tech [15]. NU-Tech is a specific platform to test and tune real-time DSP algorithms through a PC workbench. A Plug-In architecture allows to write NU-Tech Satellites (NUTSs) in C++ and plug them into the graphical interface to test the results on a common PC. The internal parameters of every Plug-In can be modified using the RTWatch (RealTime Watch). In this case the RTWatch has been used to change several parameters of the algorithm as:

- step size μ_w, μ_s, μ_h;
- number of subbands;
- length of prototype filter;
- primary path length;
- secondary path length.

Figure 2 shows the scheme used to evaluate the algorithm. The central block entitled "proposedANC" implements the algorithm proposed while the three windows show the performance of the proposed algorithm considering white noise as input. The NUTS developed with the proposed approach has three inputs, the first pin is used as input of the reference signal $x(n)$, the second pin for the desired signal $d(n)$ and the third pin for the injection noise used for the online estimation of secondary path. The outputs of the NUTS have been used to test the proposed approach and to compare with the approach of [12]. The first output pin is used to plot the residual error $e'(n)$ of primary adaptation, the second pin is used to show the residual error $e_s(n)$ of secondary adaptation and the output of the third pin is the signal $y(n)$, i.e., the control signal of the secondary source.

IV. Experimental Results

The system implementation has been firstly validated with MATLAB by testing the proposed algorithm with snoring signals and comparing it with [12]. Subsequently, the proposed algorithm has been tested with two different snoring signals downloaded from [16] and comparing it with the approach proposed by [12] using the NU-Tech software, considering a sample rate of 44.1 kHz and a frame size of 4096 samples. The paths employed to test the algorithms are measured inside a semianechoic chamber from the setup proposed in [6] where an headboard has been equipped with loudspeakers and microphones. The primary and secondary paths are stored as filters of N = 256 taps. The performance of the proposed algorithm has been evaluated considering the follows signals:

- the evaluation of primary and secondary path;
- the convergence rate of the primary path;
- the input error of the secondary path adaptation;
- the mean squared error (MSE) after cancellation.

The following equation has been employed to evaluate the convergence rate of primary path:

$$\Delta w_e(n) = \frac{||w(n) - p(n)||}{||p(n)||}, \tag{9}$$

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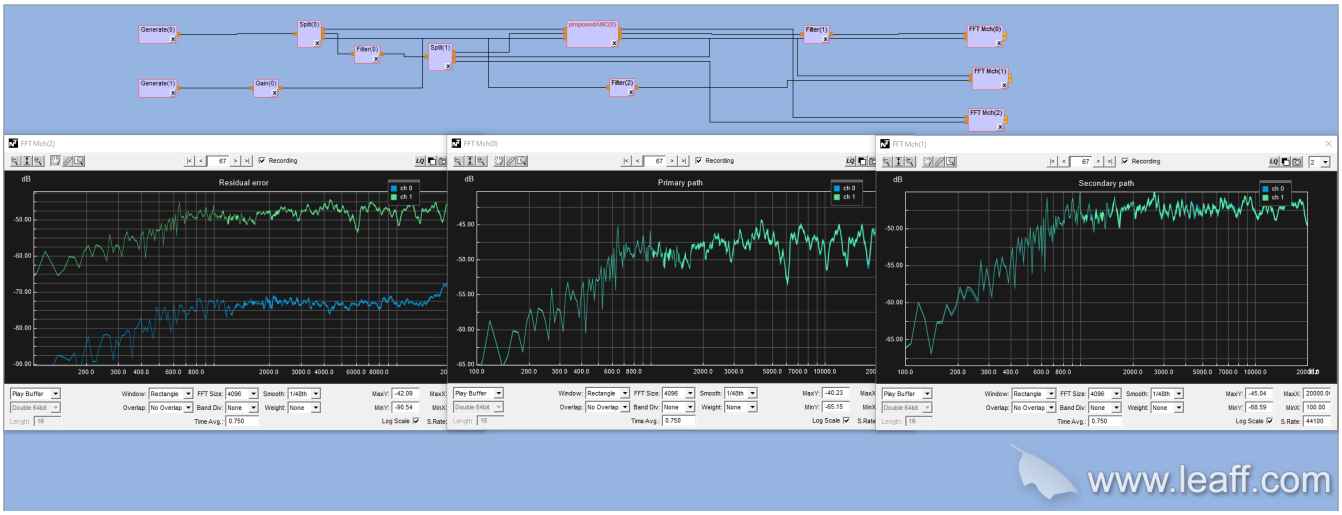


Fig. 2. Scheme of the real-time implementation.

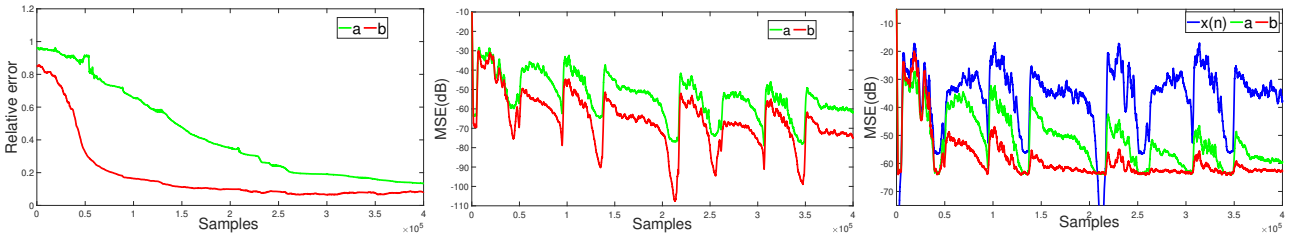


Fig. 3. Comparison between (a) the reference algorithm of [12] and (b) the proposed algorithm, evaluating (i) the primary path error estimation, (ii) the secondary path error estimation and (iii) the MSE in relation to the input signal $x(n)$, considering snoring signal as input.

where $w(n)$ describes the taps of the main adaptive algorithm and $p(n)$ represents the taps of the primary path.

A. Validation Tests

The proposed system has been firstly validated through the MATLAB software comparing its performance with the approach of [12] using white noise and snoring signals as input. The length of the adaptive filters has been chosen of 256 taps for the subband filtering, for $\hat{s}(n)$ and $h(n)$. For the delayless algorithm, a prototype filter length of $L = 256$ has been considered for the case of white noise input while a length of $L = 256$ has been considered for snoring noise input. Different simulations have been carried out to find the best values of step size and of prototype filter length of the proposed algorithm. For the reference algorithm the identified values are the following: $\mu_w = 0.026$, $\mu_s = 0.006$, $\mu_h = 0.001$. For the proposed algorithm the identified values are the following: $\mu_w = 0.061$, $\mu_s = 0.026$, $\mu_h = 0.001$. For the SAF structure, a number of subband of $M = 32$ has been selected and the length of the prototype filter is set to $L = 64$. Figure 3(i) presents the convergence rate of the primary path calculated by Equation (9) showing that the proposed algorithm exhibits a significant improvement compared to the reference approach. This improvement is confirmed also by the convergence rate of secondary path adaptation that is shown in Figure 3(ii) where the input error $e_s(n)$ is displayed.

Figure 3(iii) compares the input noise with the residual error $e(n)$ and this Figure shows as the proposed approach achieves a lower residual and a lower time of convergence error than the reference algorithm.

B. Real time Experiments

To validate the real time implementation both algorithms, i.e., proposed approach and approach of [12] have been implemented taking into account two different snoring noises as input. Even in this case, several test were carried out to find the optimal values of the step size and length of the prototype filter for the proposed algorithm. Considering the first snoring signal, the found values for the reference algorithm of the state of the art are $\mu_w = 0.026$, $\mu_s = 0.025$, $\mu_h = 0.001$ while for the proposed algorithm are $\mu_w = 0.046$, $\mu_s = 0.051$, $\mu_h = 0.01$. For the SAF structure, a number of subband of $M = 64$ and the prototype filter length $L = 128$ have been selected. Regarding the primary path estimation, both algorithms perfectly follow its the frequency response below 10 kHz as shown in Figure 4(i). Above this frequency, the reference algorithm is unable to follow the path due to a lowest

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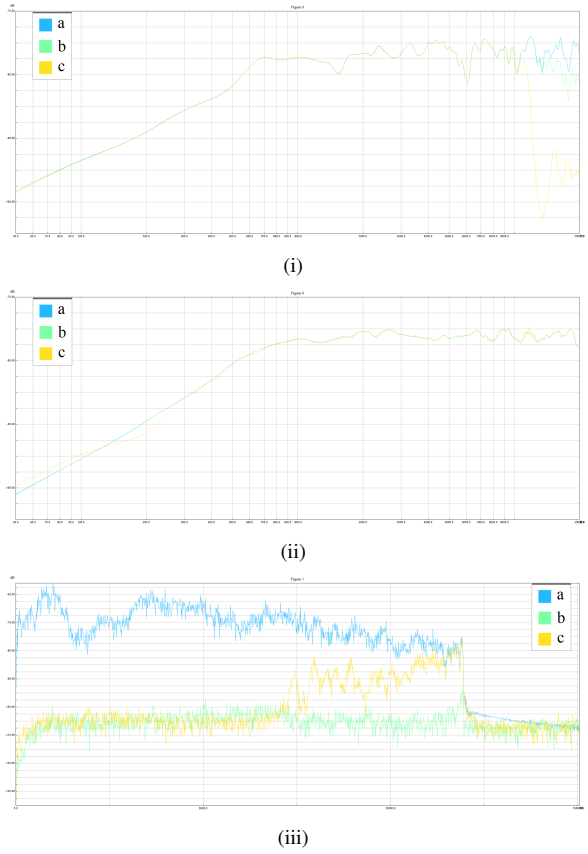


Fig. 4. Comparison between (a) paths measured, (b) the proposed adaptive filter weights and (c) the adaptive filter weights of [12] for the primary path and secondary path estimation considering snoring noise as input. The Figure below compares the residual error in frequency.

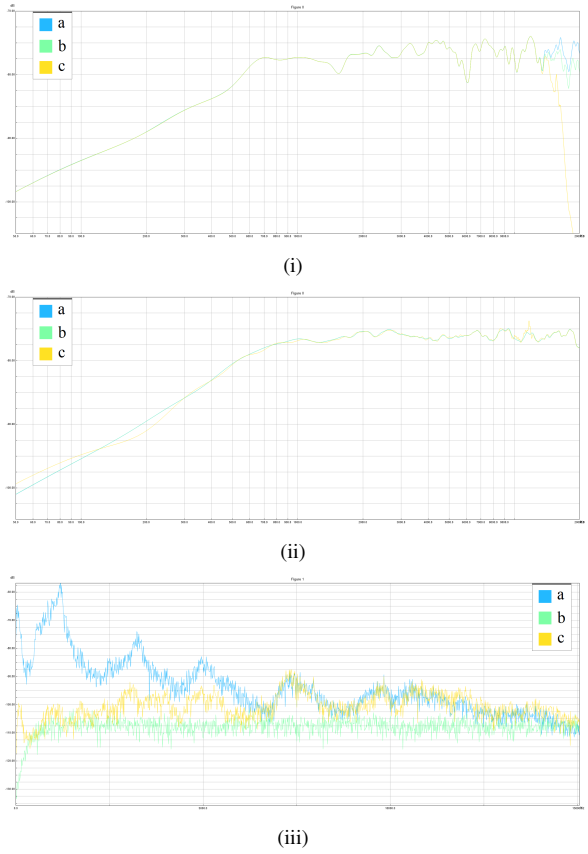


Fig. 5. Comparison between (a) paths measured, (b) the proposed adaptive filter weights and (c) the adaptive filter weights of [12] for the primary path and secondary path estimation considering snoring noise as input. The Figure below compares the residual error in frequency.

power of the input signal in that frequency range. This issue is solved by proposed algorithm that is capable of reconstructing perfectly the frequency response up to 20 kHz as shown in the experiments. Regarding the secondary path estimation, the proposed algorithm fits perfectly its response as shown in Figure 4(ii) whereas the reference algorithm deviates from the measured path in the low frequency and above 10 kHz. Also the convergence rate is improved as shown in Figure 4(iii) where the residual error signal is plotted in frequency domain after 3 seconds of adaptation. The Figure shows as the proposed method reaches the noise floor after 3 seconds of adaptation.

Considering the second snoring signal the found values for the adaptation procedure are $\mu_w = 0.031$, $\mu_s = 0.028$, $\mu_h = 0.001$ while for the proposed algorithm the identified values are $\mu_w = 0.050$, $\mu_s = 0.055$, $\mu_h = 0.01$. For the SAF structure, a number of subband of $M = 64$ and the length of the prototype filter of $L = 128$ have been selected. Regarding the primary path estimation, both algorithms perfectly follow its frequency response below 14 kHz as shown in Figure 5(i). Above this frequency, the reference algorithm is unable to follow the path as for the previous snoring signal. Regarding the secondary path estimation, the proposed algorithm fits perfectly its response as shown in Figure 5(ii) whereas the reference algorithm deviates from the measure path in the low frequency and above 10 kHz. Also the convergence rate is improved as shown in Figure 5(iii) where the residual error signal is plotted in frequency after 3 seconds of adaptation. The Figure shows as the proposed method reaches the noise floor after 3 seconds of adaptation. The convergence rate is improved even for the secondary path adaptation as shown in Figures 6(i)-6(ii) where the error $e_s(n)$ is plotted. The Figures show as the error is really low in the case of the proposed algorithm while the reference algorithm has a slower convergence and lower performance in terms of adaptation. Also the Figures 7-8 show a better convergence rate where error signal of both algorithms is calculated with the real time implementation. As shown in the Figure, the proposed algorithm has a better convergence rate and a residual noise lower than the reference approach.

V. Conclusions

In this paper, a real time implementation of new structure for active noise control that includes a secondary path adaptation algorithm and a SAF algorithm for primary path estimation is presented and employed for snoring noise cancellation. Several

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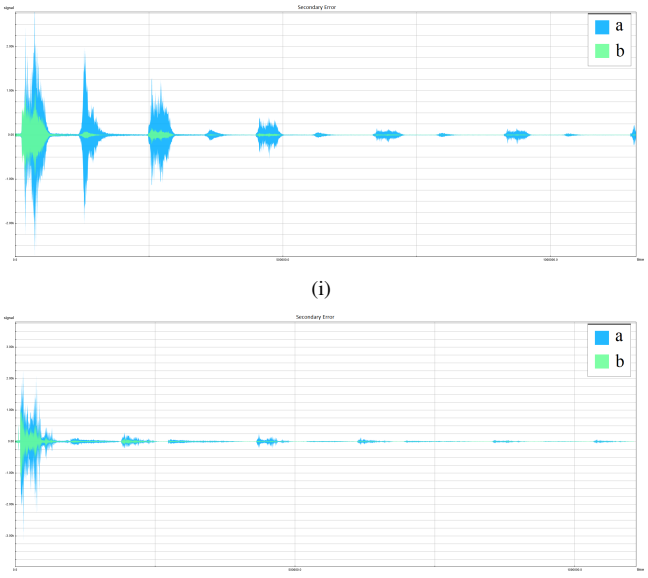


Fig. 6. Error signal $e_s(n)$ of the secondary path adaptation.

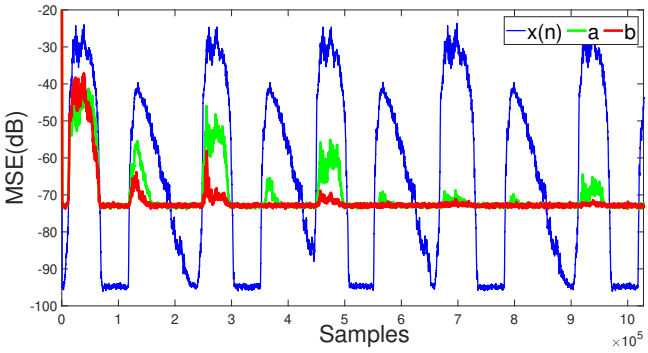


Fig. 7. MSE in relation to the first snoring signal $x(n)$.

tests have been carried out and the proposed approach has been compared with an state-of-the-art algorithm. The real time experiments have confirmed the performance of the proposed approach in particular the proposed technique have a lower error in primary path estimation and a lower time to reach the convergence for primary and secondary path. The application of the SAF technique improves the estimation in low frequency of paths in particular when noise snoring is used as input noise signal. Future works will be centered on implementation and testing of the proposed approach in a real environment.

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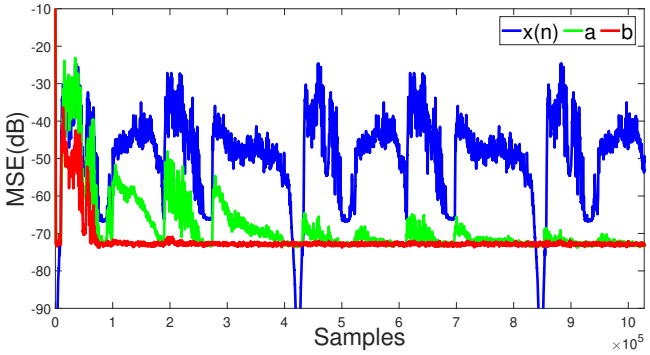


Fig. 8. MSE in relation to the second snoring signal $x(n)$.

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